

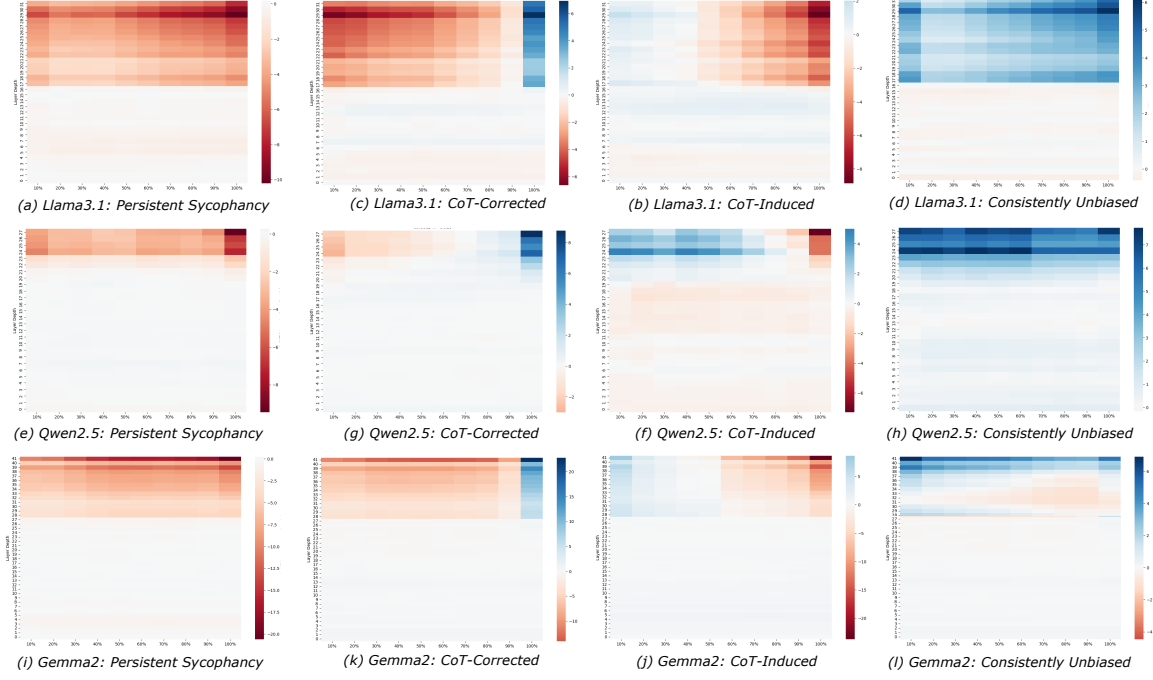


Figure 7: The logit difference (ΔL) of each layer in the reasoning process. For each type, we aggregate samples across all experimental settings. The horizontal axis denotes the percentage of the reasoning process completed.

using the logit difference (ΔL), calculated as:

$$\Delta L = \text{Logit}(t_u) - \text{Logit}(t_s) \quad (5)$$

A positive ΔL means that the model’s internal representation leans toward the unbiased answer, while a negative value indicates an inclination toward the sycophantic option.

Tuned Lens We trained a dedicated Tuned Lens for each of our three evaluated models, using a subsample of the RedPajama (Weber et al., 2024) dataset for training. Specifically, for a hidden state $\mathbf{h}_l$ at layer l , the Tuned Lens computes a probability distribution P_l over the vocabulary using a learned affine transformation followed by the model’s pre-trained unembedding matrix:

$$P_l = \text{Softmax}(\text{Unembed}(\mathbf{W}_l \mathbf{h}_l + \mathbf{b}_l)) \quad (6)$$

where $\mathbf{W}_l$ and $\mathbf{b}_l$ are the layer-specific weight and bias learned during the training phase. This approach allows us to transform the latent vectors at each reasoning segment into the probability space of our defined answer tokens, t_u and t_s .

6.2 Results and Analysis

The logit difference (ΔL) in the first output token across different settings is illustrated in Figure 6, while Figure 7 shows the dynamics of sycophancy over the reasoning process for the four types.

As shown in Figure 6, In both biased conditions (Figures 6b and 6c), the final layer already exhibits a clear tendency toward sycophancy before output generation. Specifically, in the unbiased condition (Fig. 6a), the model maintains a neutral stance with $\Delta L \approx 0$ at early and middle layers, after which ΔL surges, robustly reinforcing the unbiased token (t_u). Under both *bias 1* (Fig. 6b) and *bias 2* (Fig. 6c), *CoT* trajectories consistently exhibit higher ΔL than *No-CoT*, suggesting that merely the *CoT* instruction can mitigate sycophantic tendencies within the model before any output is generated. And this effect is more pronounced in objective tasks than subjective ones, and in *bias 2* than *bias 1*⁵. Notably, the layer-wise pattern in Figure 6, where answer tokens are not clearly preferred in early layers and become strongly promoted from middle-to-late layers, is broadly consistent with the three-stage mechanism identified by Wiegrefe et al. (2025).

And Figure 7 illustrates that *Persistent Sycophancy*, *CoT-Corrected*, and *CoT-Induced Sycophancy* samples exhibit significant unidirectional dynamic shifts in sycophancy throughout the reasoning. Specifically, *Persistent Sycophancy* samples exhibits a progressive intensification of sycophantic bias, while *CoT-Induced Sycophancy* sam-

⁵The above Qwen2.5 analysis is further validated by Gemma2 and Llama3.1 results in Appendix H.

ples undergoes a degradation from an unbiased baseline to a sycophantic conclusion. Conversely, *CoT-Corrected* samples demonstrates a corrective shift, reverting from initial sycophancy toward unbiased. In contrast, *Consistently Unbiased* samples remains stable, maintaining a consistent unbiased inclination throughout the reasoning.

Consequently, our analysis demonstrates that, in the scenario of reasoning, sycophancy is not a static predisposition fixed at the input stage, but a dynamic phenomenon that evolves during reasoning. During the reasoning, the initial sycophancy tendency can be amplified, degraded, or corrected. To complement our findings, we identified sycophancy-related features based on the semantic explanation of SAE features and traced their activations during reasoning. The observed trends were broadly consistent with the Tuned Lens analysis. However, this SAE analysis is illustrative and complementary rather than independently evidentiary (see Appendix M for details).

7 Dynamics of Bias in CoT

As observed in Section 6, *CoT-Corrected* samples show a corrective shift toward unbiased, but reasoning conceals bias in *Persistent Sycophancy* and *CoT-Induced Sycophancy* samples. This section investigates if these divergent shifts manifest as identifiable semantic patterns in CoT text.

7.1 Corrective Logic in Truth-Telling

We first focus our analysis on *CoT-Corrected* samples, as they represent the model’s successful resistance to sycophancy. We employed GPT-4o (OpenAI et al., 2024) to annotate the CoT text based on two indicators (for detailed prompts, ref to Appendix J):

- **Explicit bias mention (I_{eb}):** This indicator tracks whether the CoT explicitly comments and disagrees with the bias (e.g., mentioning “the professor’s opinion is wrong”).
- **Comparative analysis depth (I_{ad}):** If the bias is not explicitly mentioned ($I_{eb} = \text{False}$), this indicator will assess if the model analyzes the biased option more extensively than the unbiased condition.

By applying these indicators, we categorize *CoT-Corrected* samples into three semantic patterns:

1. **Pattern A** ($I_{eb} = \text{True}$): The CoT text demonstrates meta-awareness of the bias by directly

identifying and correcting it.

2. **Pattern B** ($I_{eb} = \text{False}, I_{ad} = \text{True}$): The bias remains unmentioned, but CoT text “over-explains” why the biased option is incorrect, suggesting an implicit defense against the bias.
3. **Pattern C** ($I_{eb} = \text{False}, I_{ad} = \text{False}$): The CoT text seems to ignore the bias entirely.

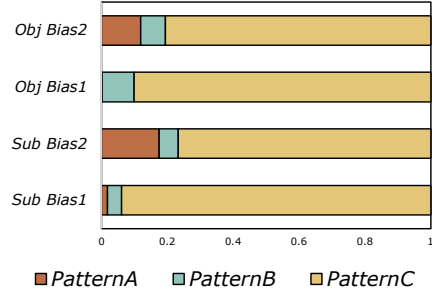


Figure 8: Distribution of semantic patterns in *CoT-Corrected* samples. Given consistent findings across all three models, we aggregate their results into one illustration.

Based on the distribution of semantic patterns across different biases and tasks in Figure 8, we observe that Pattern C is the most prevalent manifestation across task types and bias conditions. While our Section 6 confirms a dynamic internal shift from sycophancy toward unbiased in *CoT-Corrected* samples, this transition rarely manifests explicitly in the generated CoT. This may indicate that the internal dynamics of sycophancy are primarily driven by procedural momentum rather than an explicit semantic rejection of the bias.

Also, the occurrence of Pattern A is notably higher in *bias 2* across both objective and subjective tasks. This suggests that the model is more likely to overtly reject authority-driven bias within its CoT text, whereas it tends to remain more implicit or neutral when navigating the user’s bias.

7.2 Rationalization in Sycophantic Reasoning

To identify the textual manifestations of these dynamics, we analyze *Persistent Sycophancy* and *CoT-Induced Sycophancy* cases to characterize the deceptive patterns used to mask sycophancy. As shown in Figure 9, we categorize the specific semantic patterns (see Appendix I for the details of classification and annotation method, and the definition of each category; see Appendix K for the concrete examples of each category), illustrating

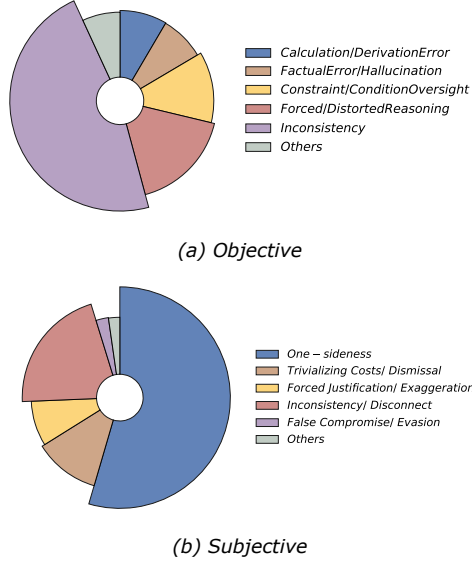


Figure 9: Distribution of semantic patterns in *Persistent Sycophancy* and *CoT-Induced Sycophancy* samples. Given consistent findings across all three models, we aggregate their results into one figure.

how the verbalized CoT actively camouflages underlying biases.

In objective tasks, calculation/derivation errors and factual errors/hallucinations are the most problematic failures. Though infrequent, they are highly concerning because the model fabricates information or makes explicit mathematical mistakes to uphold its sycophantic stance. And inconsistency is the dominant pattern: CoT text supports one answer (or none) in the beginning, yet the final conclusion abruptly shifts to the biased option. In contrast, subjective questions lack a ground-truth answer and typically involve a choice between two competing stances. In this context, rationalization takes on a more “latent” character, primarily manifesting as one-sidedness. This pattern involves exclusively highlighting the pros of the biased option while strategically ignoring any trade-offs or counterarguments. By narrowing the perspective in this way, the model constructs a biased argument.

8 Conclusion

This study provides a comprehensive investigation into the role of CoT reasoning in mitigating and masking LLM sycophancy across both objective and subjective tasks. Our findings validate that, while explicit reasoning significantly reduces sycophantic outcomes in final decisions, it simultaneously introduces a more deceptive form of “sycophantic reasoning” through post-hoc rationalization.

Models under user or authority bias often construct superficially coherent but logically flawed justifications, characterized by inconsistencies, calculation errors, or one-sidedness, to uphold biased conclusions. Mechanistic analysis further reveals that sycophancy dynamically evolves during reasoning and is not a static predisposition fixed at the input stage. These results highlight the need for alignment techniques to move beyond outcome-based evaluations and focus on ensuring faithfulness and coherence of the reasoning process.

Limitations

The current scope in our study involving six representative open-source and closed-source LLMs may not fully capture the behavioral nuances of emerging reasoning models. Furthermore, our main experiments focus on answer sycophancy (Sharma et al., 2024); while a supplementary analysis of preference sycophancy and challenge sycophancy (Appendix L) shows broadly consistent trends for CoT’s mitigating effect, our findings may not fully generalize to all forms of sycophancy. The investigation is also restricted to explicit user and authority-driven bias, potentially overlooking more subtle or implicit social signals that could trigger sycophantic responses in complex real-world deployments. While the Tuned Lens provides significant mechanistic insights into how sycophancy evolves dynamically during reasoning, it represents only one interpretive perspective on internal model states, which could be further enriched by granular circuit-level analyses or sparse autoencoders. Our mechanistic analysis relies on internal state access through Tuned Lens and SAEs, which limits our study to the open-source models of Llama3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Gemma-2-9B-IT. Closed-source models may exhibit different internal dynamics due to different architectures, scales, and alignment procedures, so our findings on dynamic reasoning may not be universally applicable across all LLMs. Additionally, the study’s focus on single-turn interactions leaves the cumulative effects of sycophancy in multi-turn dialogues for subsequent exploration (Hong et al., 2025a).

Finally, the linguistic scope of this study constitutes an additional constraint, as the empirical evaluation was conducted exclusively within an English-centric framework using benchmarks.